FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Advanced)-2019

PAPER – 2

TEST DATE: 13-05-2019

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01 – 06, 19 – 24, 37 – 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 – 46) *contains 12 questions. Each question has* ***TWO (02)*** *matching lists:* ***LIST-I*** *and* ***LIST-II****.* ***FOUR*** *options are given representing matching of elements from* ***LIST-I*** *and* ***LIST-II****.* ***ONLY ONE*** *of these four options corresponds to a correct matching.*
   For each question, choose the option corresponding to the correct matching.
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A light rod of length ℓ is free to rotate about one of its end in vertical plane. A ring of mass m can slide freely without friction on the rod. Initially rod is kept horizontal and ring is situated at a distance $3\ell/5$ from fixed end. Now, rod is released choose correct statement(s).

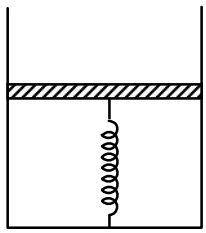
   (A) Velocity of ring when it leaves contact with rod is $\sqrt{\frac{8g\ell}{5}}$.

   (B) Angular momentum of the ring about point of suspension of rod when it looses contact with rod is $m\sqrt{\frac{72g\ell^3}{125}}$.

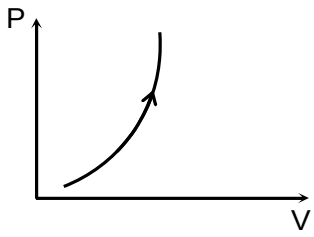
   (C) Angular velocity of the rod remains constant through out the motion.

   (D) Acceleration of the ring is uniform.

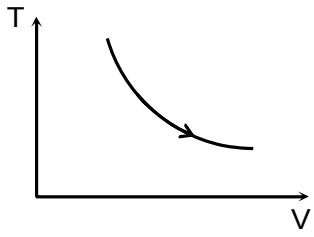
2. A piston of negligible mass is fitted with a cylinder and connected to spring of spring constant k as shown in figure. An ideal mono-atomic gas is filled in cylinder and piston is in equilibrium. Now gas in heated slowly. Which of the following graph(s) can be correct representation of process? (Initial pressure of gas is greater than atmospheric pressure).



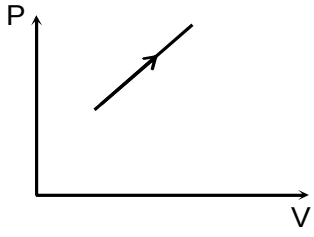
(A)



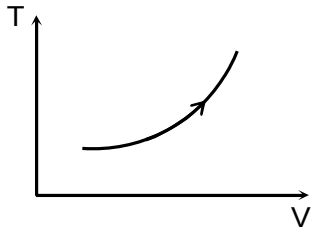
(B)



(C)

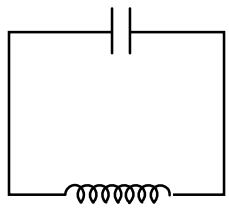


(D)



3. A car is moving towards a stationary observer with speed 10 m/s. The car blows a horn of frequency 640 Hz for 10 seconds. If speed of sound in air is 330 m/s, then
   (A) Frequency of horn heard by stationary observer is 660 Hz
   (B) The time duration during which observer hears the horn is 12 s.
   (C) Frequency of horn heard by driver of car is 640 Hz
   (D) The time duration during which observer hears the horn is 10 s.

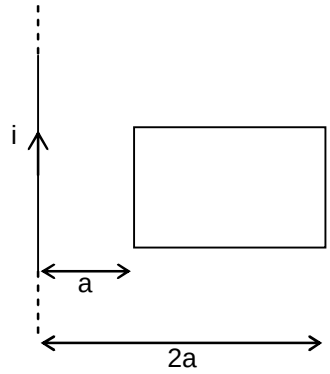
4. In an LC circuit shown in figure C = 2F and L = 2H. At t = 0 charge on capacitor is $4\sqrt{3}\ \mu C$ and at same instant is increasing at a rate of 6 μC/s. Choose the correct statement(s):



   (A) Maximum charge on capacitor $8\sqrt{3}\ \mu C$.
   (B) Maximum current through inductor is $4\sqrt{3}\ \mu A$.
   (C) Total energy stored in circuit 48 μJ.
   (D) Total energy stored in circuit is 24 μJ.

5. A ring of radius R is made up of thin wire of linear mass density ρ having cross section area A and Young's modulus Y is made to rotate about its axis with angular velocity ω then,
   (A) Tension in ring $\rho\omega^2R^2$
   (B) Elastic potential energy stored in ring is $\frac{2\pi\rho^2\omega^4R^5}{YA}$
   (C) Tension in ring is $\frac{\rho}{2}\omega^2R^2$
   (D) Kinetic energy of the ring is $\pi\rho R^3\omega^2$

6. An infinitely long straight wire is carrying a constant current i. There is a square closed loop made of conducting wire which is placed near the wire as shown in figure. Choose correct statement(s) for above situation.

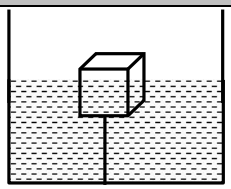
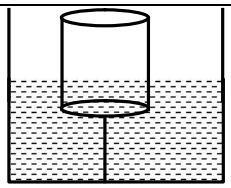
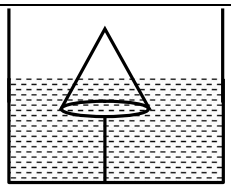
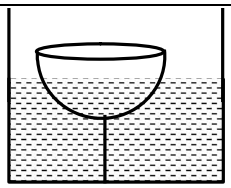


   (A) If loop is taken away from wire the current in the loop will be clockwise.
   (B) Mutual inductance of the system for the situation shown in figure will be $\frac{\mu_0 a}{4\pi}\ln 2$
   (C) If loop is taken away from the wire the current in the loop will be anticlockwise.
   (D) Mutual inductance of the system for the situation shown in figure will be $\frac{\mu_0 a \ln 2}{2\pi}$

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

7. Different geometrical constructions of density ρ are tied to string such that they are floating in a liquid of density 3ρ with half of their volume outside liquid as shown in List –I and the value of tension in the string is given in List-II.

LIST–I		LIST–II	
P.	 **A cube of side 2a**	1.	$\pi a^3 \rho g$
Q.	 **A cylinder of height 2a and radius a**	2.	$4a^3 \rho g$
R.	 **A cone of height 3a and base radius a**	3.	$9\pi a^3 \rho g$
S.	 **A solid hemisphere of radius 3a**	4.	$\frac{\pi a^3 \rho g}{2}$

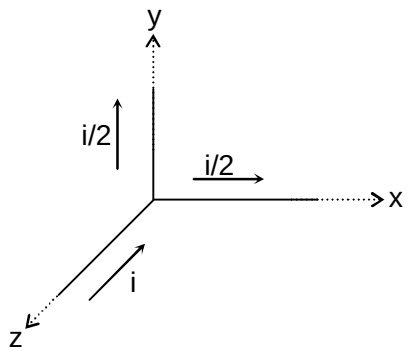
(A) P → 4; Q → 1; R → 3; S → 2

(B) P → 2; Q → 1; R → 4; S → 2

(C) P → 3; Q → 4; R → 1; S → 2

(D) P → 2; Q → 1; R → 4; S → 3

8. Three long wire carrying current are placed along axis. The magnitude and directions of current are also shown in figure. List- I contains the coordinates of certain points and List-II contains magnetic field at points listed in List-I, not necessarily in same order.



	LIST–I		LIST–II
P.	(–2a, 0, 0)	1.	$\frac{\mu_0 i}{8\pi a}(-\hat{i})+\frac{\mu_0 i}{16\pi a}(-\hat{k})$
Q.	(0, –a, 0)	2.	$\frac{\mu_0 i}{4\pi a}(-\hat{i})+\frac{\mu_0 i}{8\pi a}(-\hat{k})$
R.	(0, 0, –4a)	3.	$\frac{\mu_0 i}{16\pi a}(\hat{k})+\frac{\mu_0 i}{8\pi a}(\hat{j})$
S.	(0, –2a, 0)	4.	$\frac{\mu_0 i}{32\pi a}(-\hat{i})+\frac{\mu_0 i}{32\pi a}(\hat{j})$

(A) P → 4; Q → 1; R → 3; S → 2
(B) P → 3; Q → 2; R → 4; S → 1
(C) P → 3; Q → 4; R → 1; S → 2
(D) P → 4; Q → 1; R → 2; S → 3

9. Electric field in a region is given by $\vec{E}=\left[(2xy)\hat{i}+(x^2)\hat{j}\right]$ V/m. Electric potential at origin is 5V. Match the following two columns.

	LIST–I		LIST–II
P.	Potential at point (5 m, 2m, 1m)	1.	4 V
Q.	Potential at point (1m, 1m, 1m)	2.	1 V
R.	Potential at point (2m, 1m, 1m)	3.	3 V
S.	Potential at point (1m, 2m, 1m)	4.	–45 V

(A) P → 4; Q → 1; R → 3; S → 2
(B) P → 3; Q → 1; R → 4; S → 2
(C) P → 3; Q → 4; R → 1; S → 2
(D) P → 4; Q → 1; R → 2; S → 3

10. In a YDSE setup a transparent film of refractive index 1.5 is placed on one of the slit. Different values of t are given in column I. Column II contains the ratio of intensity at central line on screen after inserting the slab and before inserting the slab. Wavelength of light used in experiment is 600 nm. Then

	LIST–I		LIST–II
P.	400 nm	1.	3/4
Q.	200 nm	2.	1/4
R.	300 nm	3.	1/2
S.	600 nm	4.	0

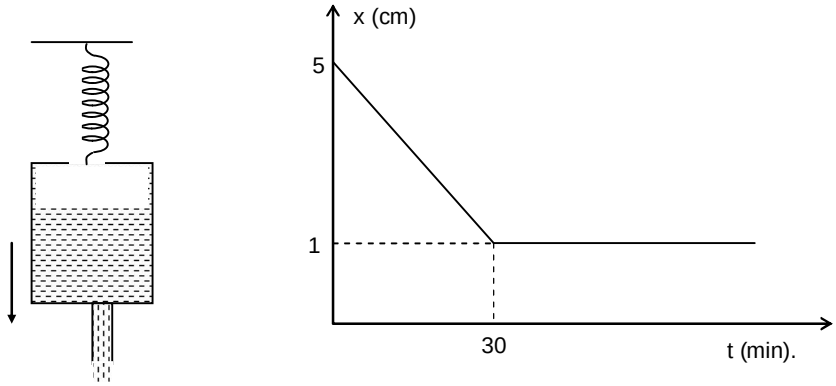
(A) P → 4; Q → 1; R → 3; S → 2

(B) P → 2; Q → 1; R → 3; S → 4
(C) P → 3; Q → 4; R → 1; S → 2
(D) P → 2; Q → 1; R → 2; S → 3

SECTION – D
(Numerical Answer Type)

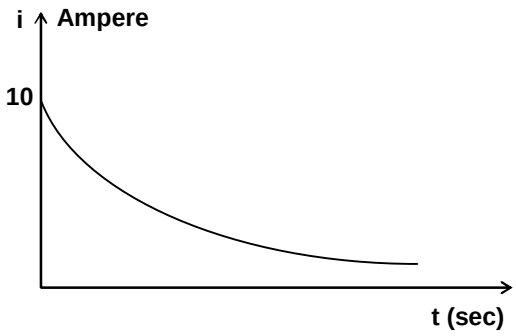
This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. A container filled with water is suspended with the help of a spring of spring constant 500 N/cm. A small orifice is made in container and water starts flowing out. The variation of elongation in spring with time is shown in graph calculate the flow rate of water in cm^3/sec. [Neglect any oscillations]



12. A circular plane mirror is kept on a horizontal surface, an opaque sphere of radius R = 20 cm is placed symmetrically on the circular plane mirror. An insect lies on the sphere at a vertical height of 36 cm from the circular mirror. Find the minimum radius of mirror so that image is formed by the mirror.

13. An R-C series circuit having resistance R = 100 Ω is switched on at t = 0. The graph of current v/s time is shown in figure. The value of initial potential difference across capacitor is (in Volts)



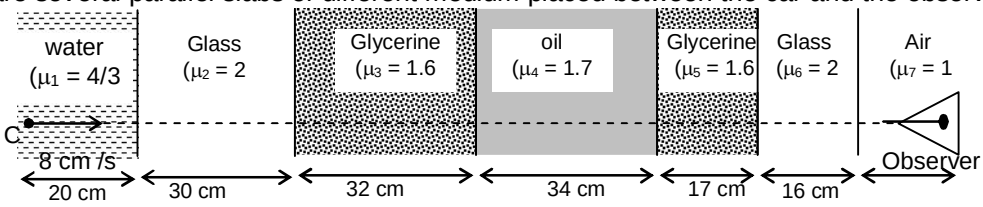
14. An open organ pipe is vibrating in air and it is vibrating in 3rd overtone with maximum pressure amplitude $\sqrt{3}$ N/m^{-2} the value of maximum volumetric strain at a distance (1/12)m from any end is $k \times 10^{-5}$. The value of k is (Adiabatic Bulk modulus of air = 100 kN/m^2, length of organ pipe = 1m)

15. Total mass of a system of particles is 20 kg. To apply Newton's law from the centre of mass frame on a particle of mass 5 kg one has to apply pseudo force $\vec{F} = (10\hat{i} + 20\hat{j})$N on it. The magnitude of net force acting on system of particle in Newton is F_1. The value of F_1^2 is

16. What will be mean life (in hours) of a radioactive sample, if it is known that its activity decrease at the rate of 19 % per hour? [Given ln(0.9) = –0.10].

17. A man can jump over 4 m wide trench on earth. Mean density of an imaginary planet is twice that of earth, calculate maximum possible radius of that planet (in m) so that same man can escape from planet by jumping. (Radius of earth = 6400 km, use $\sqrt{10}$ = 3.16)

18. Find the velocity (in cm/s) of the car C as appeared by the observer shown in the figure. There are several parallel slabs of different medium placed between the car and the observer.



Chemistry

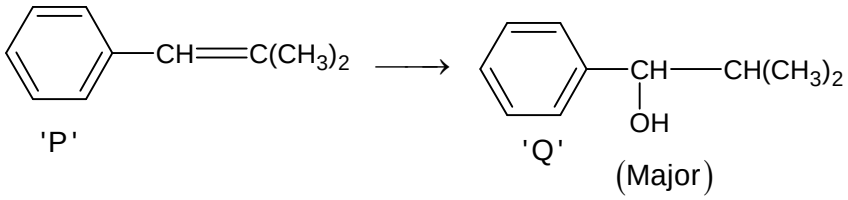
PART – II

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

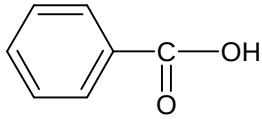
19. Which of the following statement(s) is/are correct regarding Cu^{2+} ion and Cd^{2+} ion (M = Cu or Cd)
    (A) Only CdS is soluble in yellow ammonium sulphide.
    (B) Only Cu^{2+} ion form $K_3[M(CN)_4]$ complex when treated with excess of KCN.
    (C) Only Cd^{2+} ion form $[M(NH_3)_4]^{2+}$ complex when treated with excess of NH_3.
    (D) Cu^{2+} ion gives back ppt. of CuS with H_2S gas in presence of HCl.

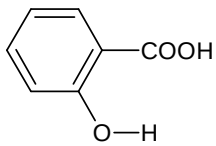
20. Which of the following reagent(s) is/are correct regarding the following conversion of (P) into (Q)

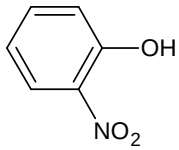


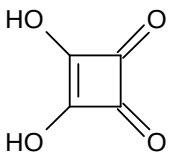
'P' 'Q' (Major)

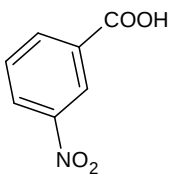
    (A) (i) Sia_2B – H, THF
        (ii) H_2O_2 - OH^-
    (B) (i) $Hg(OAc)_2$ – THF – CH_3OH
        (ii) $NaBH_4/OH^-$
    (C) (i) Conc. H_2SO_4
        (ii) H_2O
    (D) (i) HBr, peroxide
        (ii) Aq. KOH

21. Which of the following is/are stronger acid than 

    (A) 

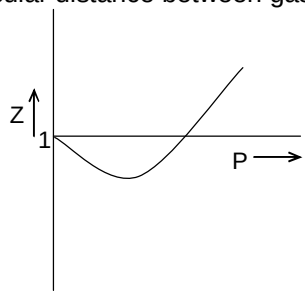
    (B) 

    (C) 

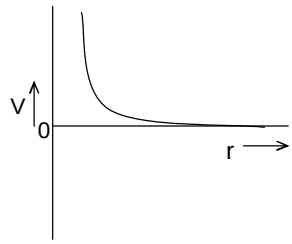
    (D) 

22. If intermolecular force of attraction between the atoms of a real gas is zero, which of the following graphs is/are correct regarding that gas (Z = compressibility factor, V = Potential energy, r = intermolecular distance between gas particle)

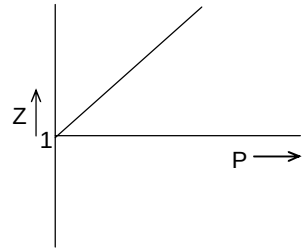
(A)



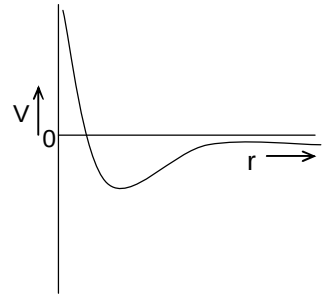
(B)



(C)

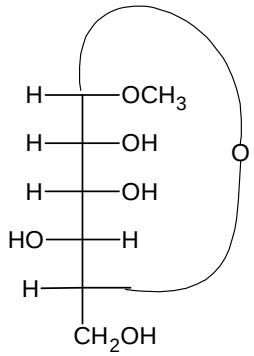


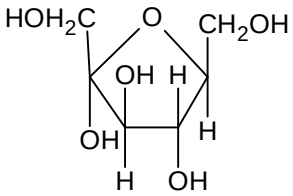
(D)

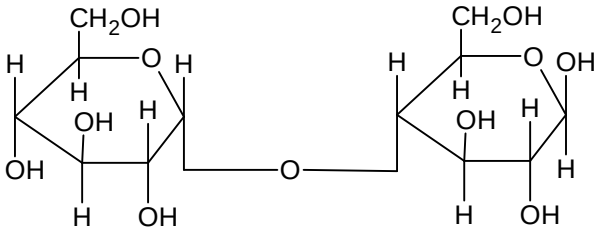


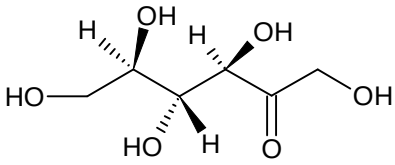
23. Which of the following is/are reducing sugar?

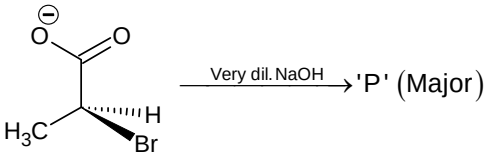
(A)



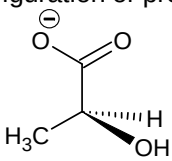
(B) 

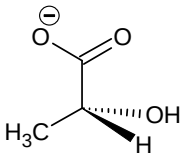
(C) 

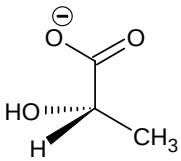
(D) 

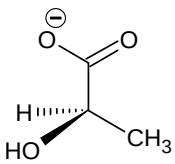
24. 

The configuration of product 'P' is/are:

(A) 

(B) 

(C) 

(D) 

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

25. Match each coordination compound in **List – I** with an appropriate pair of characteristics from **List – II** and select the correct answer using the code given below the list [Atomic number Co = 27, Cr = 24, Ni = 28, Pt = 78, Ox = oxalate ion]:

List – I		List – II	
P.	$[Co(NH_3)_4(H_2O)Cl]Br$	1.	sp^3d^2 hybridization
Q.	$K_3[Cr(Ox)_2Cl_2]$	2.	Geometrical isomerism
R.	$[Ni(H_2O)_6]Cl_2$	3.	Optical isomerism
S.	$[Pt(NH_3)_2Cl_2]$	4.	Ionization isomerism
		5.	Diamagnetic
		6.	dsp^2

The correct option is

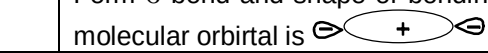
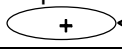
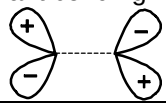
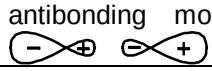
(A) P $\rightarrow$ 1, 2; Q $\rightarrow$ 2, 5; R $\rightarrow$ 1, 4; S $\rightarrow$ 2, 6

(B) P $\rightarrow$ 1, 2, 4; Q $\rightarrow$ 2, 3; R $\rightarrow$ 1; S $\rightarrow$ 2, 5, 6

(C) P $\rightarrow$ 2; Q $\rightarrow$ 2, 3, 4; R $\rightarrow$ 4, 5; S $\rightarrow$ 2, 3

(D) P $\rightarrow$ 4, 5, Q $\rightarrow$ 1, 5; R $\rightarrow$ 4; S $\rightarrow$ 2, 5

26. Atomic orbitals of two elements combine to form molecular orbitals. The axis which connect the nucleus of two elements is y axis then match each sets of atomic orbitals of **List – I** with **List – II** showing shape of molecular orbital.

List – I		List – II	
P.	$p_z + p_z$	1.	Form σ bond and shape of bonding molecular orbital is 
Q.	$p_y + p_y$	2.	Form σ bond and shape of bonding molecular orbital is 
R.	$p_x + p_y$	3.	Form π bond and shape of antibonding molecular orbital is 
S.	$s + p_y$	4.	Don't form any molecular orbital
		5.	Form σ bond and shape of antibonding molecular orbital is 

The correct option is

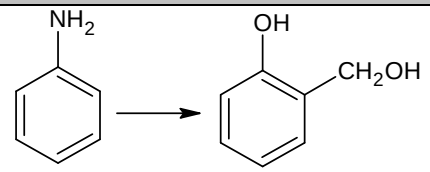
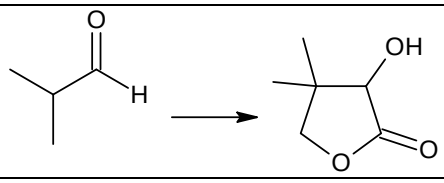
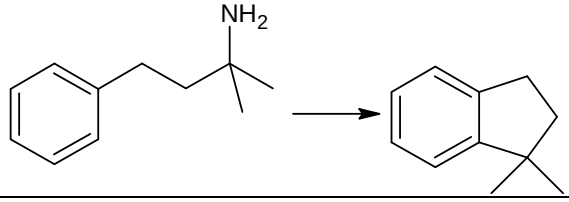
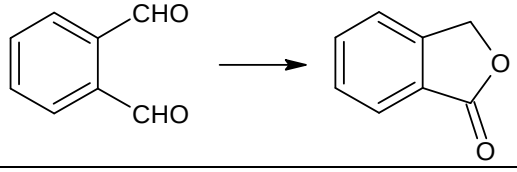
(A) P $\rightarrow$ 1, 5; Q $\rightarrow$ 3; R $\rightarrow$ 2, 5; S $\rightarrow$ 4

(B) P $\rightarrow$ 5; Q $\rightarrow$ 1, 2; R $\rightarrow$ 4; S $\rightarrow$ 5

(C) P $\rightarrow$ 3; Q $\rightarrow$ 1, 5; R $\rightarrow$ 4; S $\rightarrow$ 2

(D) P $\rightarrow$ 2, 5, Q $\rightarrow$ 3; R $\rightarrow$ 1, 2; S $\rightarrow$ 4

27. Match the chemical conversions in **List – I** with appropriate reagent(s) which is/are required for conversion in **List-II** and select the correct answer using the code given below:

List – I		List – II (Reagents)	
P.		1.	(i) Oleum (ii) NaOH fused (iii) H^+
Q.		2.	$HCHO / OH^-$
R.		3.	(i) HCN / OH^- (ii) H^+, Δ
S.		4.	(i) OH^- (ii) H^+, Δ
		5.	(i) $NaNO_2$|HCl (0 – 5°C) (ii) H_2O

The correct option is

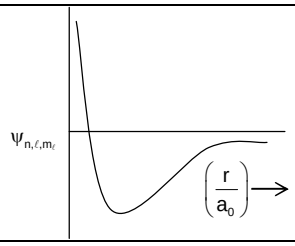
(A) P → 1, 3; Q → 4, 5; R → 1; S → 3

(B) P → 2, 5; Q → 2, 3; R → 5; S → 4

(C) P → 2; Q → 3, 4; R → 4; S → 1, 3

(D) P → 2, 5; Q → 3; R → 1, 2; S → 4, 5

28. Match each set of orbitals from **List-I** with their properties given in **List-II** (ψ_{n,ℓ,m_ℓ} is wave function depends upon spherical polar coordinate (r, θ, ϕ) of electron, z = atomic number, a_0 = Bohr's radius)

List – I		List – II	
P.	$3p_z$	1.	One nodal surface
Q.	$3d_{z^2}$	2.	Probability density is maximum at nucleus.
R.	2s	3.	$\psi_{n,\ell,m_\ell} \propto \left(\frac{Z}{a_0}\right)^{3/2} r^2 e^{-\left(\frac{Zr}{3a_0}\right)} \left(3\cos^2\theta - 1\right)$
S.	1s	4.	
		5.	$\psi_{n,\ell,m_\ell} \propto \left(\frac{Z}{a_0}\right)^{3/2} e^{-\frac{Zr}{3a_0}}$

The correct option is

(A) $P \rightarrow 3;\ Q \rightarrow 2, 3;\ R \rightarrow 4;\ S \rightarrow 2, 5$

(B) $P \rightarrow 1;\ Q \rightarrow 3;\ R \rightarrow 1, 2, 4;\ S \rightarrow 2, 5$

(C) $P \rightarrow 5;\ Q \rightarrow 1, 3;\ R \rightarrow 1, 5;\ S \rightarrow 2, 3$

(D) $P \rightarrow 1, 3,\ Q \rightarrow 3;\ R \rightarrow 1, 4, 5;\ S \rightarrow 2, 5$

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. The total number of species which having at least two unpaired electrons is X, then $\frac{X}{2}$ is:

$\left[Ni(NH_3)_6\right]^{2+}, \left[Ni(CN)_4\right]^{2-}, [PtCl_4]^{2-}, O_2, S, NO, C_2, \left[Co(NH_3)_4(NO_3)_2\right]NO_3$

$\left[Ti(H_2O)_5Cl\right]NO_3, S_2(\text{vapour phase}), \left[Mn(H_2O)_6\right]^{2+}, Se$

(Atomic number Ti = 22, Co = 27, S = 16, Ni = 28, Pt = 78, O = 8, N = 7, Se = 34 and Mn = 25)

30. A tetrapeptide made up of natural amino acid has Leucine as the N-terminal residue which is coupled to a chiral amino acid upon complete hydrolysis, the tetrapeptide gives alanine, phenylalanine, glycine and leucine. The number of possible sequences of the tetrapeptide is P, $\frac{P}{5}$ is:

31. Consider an electrochemical cell $A(s)\,|\,A^{n+}(aq, 2M)\,||\,B^{2n+}(aq, 1M)\,|\,B(s)$.

The value of standard free energy change of the reaction is – 9.5 kJ per mole of formation of B at 300 K and the EMF of the cell is $\frac{10^3}{F}$ volt. The value of n/2 is:

(Given RT = 2500 J/mol, $\ell n 2 = 0.7$, F is Faraday charge)

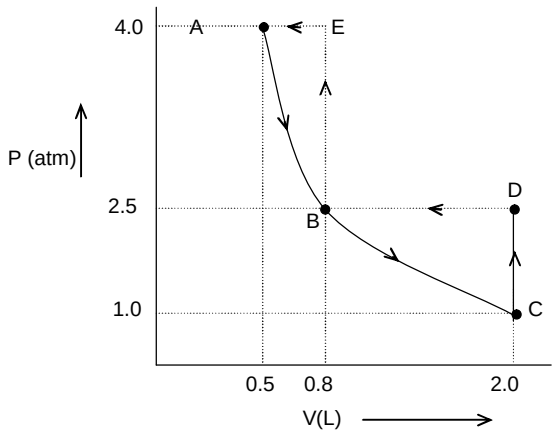
32. For the reaction, at 4 atm and 300 K equilibrium established.

$A_3B_6(g) \rightleftharpoons 3AB_2(g)$

The vapour density of equilibrium mixture is half of the vapour density of A_3B_6. If the $\Delta_r G^o$ of the system is – x Joule per mole of the dissociation of A_3B_6. what is x. [Given RT = 2500 J mol^{-1}, $\ell n 3 = 1.1$]

33. 3L of a mixture of O_2 and O_3 having pressure 8.21 atm at 27°C is treated with 800 ml of 2M KI solution buffered with a borate buffer of pH 9.2, the iodine so liberated is titrated with 200 ml of 2M hypo solution. The mole % of ozone in the mixture (Assuming ideal behaviour of gaseous mixure and there is no reaction between oxygen and hypo solution) [Given: R = 0.0821 L atm mol^{-1} K^{-1}]

34. One mole of an ideal gas is taken reversibly from state A to C through path ABC as shown below. If the gas return back to state A through the path CDBEA as shown in the figure, the net heat given by the gas (in Joule) in complete cycle is (Only magnitude of heat change) [Given: 1 L atm = 100 J, $\ell n2 = 0.7$]



35. At 300 K the half life for first order decomposition of H_2O_2 is 180 minutes. If 6 moles of H_2O_2 is left for 360 minute at 300 K, what will be the volume of O_2 at S.T.P. (in litre) produced?

36. The total number of compounds given below can discharge the pink colour of acidic solution of $KMnO_4$ is x, then $\frac{x}{5}$.

Mohr's salt, Glauber's salt, Oxalic acid, Ferrous oxalate, Sodium nitrite, Sodium thiosulphate, Ethyl alcohol, Sodium sulphates, Hydrogen sulphide, Stannous chloride.

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. Consider $S = \sum_{r=1}^{n} r\sqrt{{}^{n}C_{r}}$, where n and r are natural numbers, then which of the following statements is/are always true

(A) $S \geq 2^{n/2} n$

(B) $S \geq \sqrt{2^{n-1} n^3}$

(C) $S \leq 2^{n/2} n$

(D) $S \leq \sqrt{2^{n-1} n^3}$

38. Let $a_1, a_2, ..., a_n$ represent real numbers such that $\sum_{i=1}^{n} a_i = 1$ and $a_i \geq 0 \ \forall \ i \in \{1, 2, ..., n\}$.

If $S = \sum_{i=1}^{n-1} a_i a_{i+1}$, then S can take which of the following values

(A) 0

(B) $\frac{1}{4}$

(C) $\frac{2}{3}$

(D) 1

39. An equilateral triangle is constructed such that the origin is one of its vertices while the other two vertices lie on the line $\frac{x-3}{2} = \frac{y-2}{3} = z-2$, then which of the following statements is/are correct

(A) area of the equilateral triangle is $\sqrt{3}$

(B) inradius of the equilateral triangle is $\frac{1}{\sqrt{3}}$

(C) circumradius of the equilateral triangle is $\frac{2}{\sqrt{3}}$

(D) centroid of the equilateral triangle is $\left(\frac{2}{\sqrt{3}}, \frac{-2}{\sqrt{3}}, \frac{2}{\sqrt{3}}\right)$

40. In ΔABC, if AD is a median and AD is perpendicular to AC, then which of the following statements is/are correct

(A) $\cos A \cos C = \frac{2(c^2 - a^2)}{3ac}$

(B) $\cos A \cos C = \frac{b^2 + c^2 - a^2}{ac}$

(C) $\tan A + 2 \tan B = 0$

(D) $\tan A + 2 \tan C = 0$

41. If f(x) satisfies the differential equation $xf'(x) = f(x) + x^2 + 9(f(x))^2$ and f(1) = 0, then which of the following statements is/are true

(A) $\lim\limits_{x\to 1^+} \dfrac{f(x)}{x-1} = 0$

(B) $\lim\limits_{x\to 1^-} \dfrac{f(x)}{x-1} = 1$

(C) $\lim\limits_{x\to 0^+} \dfrac{f(1+x)+f(1-x)}{x^2} = 0$

(D) $\lim\limits_{x\to 0^-} \dfrac{f(1+x)+f(1-x)}{x^2} = 2$

42. Consider $I_n = \int_0^{\infty} \dfrac{dx}{\left(x+\sqrt{1+x^2}\right)^n}$ where n > 1, then which of the following statements is/are true

(A) $I_2 = \dfrac{2}{3}$

(B) $I_2 = \dfrac{2}{5}$

(C) $I_{10} < I_5$

(D) $I_{10} > I_5$

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

43. Let $f_1(x) = x\tan^{-1}x$, $f_2(x) = \begin{cases} \dfrac{x}{\ln 2} & ; \text{ if } x \neq 1 \\ 1 & ; \text{ if } x = 1 \end{cases}$, $f_3(x) = \begin{cases} (x+1)e^{-\left(\frac{1}{|x|}+\frac{1}{x}\right)} & ; \text{ if } x \neq 0 \\ 0 & ; \text{ if } x = 0 \end{cases}$ and

$f_4(x) = \dfrac{2\tan\left(\frac{\pi}{4}-x\right)}{\cot 2x}$ if $x \neq \dfrac{\pi}{4}$

LIST - I		LIST - II	
P.	Number of critical points for $f_1(x)$ over its domain is	1.	–1
Q.	Derivative of $f_2(x)$ at x = 1 is	2.	0
R.	Number of points of discontinuity for $f_3(x)$ in the interval [–2, 2] is	3.	1
S.	Value of $f_4\left(\frac{\pi}{4}\right)$ such that the function is continuous everywhere on interval $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$ is	4.	2
		5.	3
		6.	none of these

The correct option is:
(A) P → 4; Q → 6; R → 2; S → 2
(B) P → 3; Q → 6; R → 5; S → 3
(C) P → 3; Q → 6; R → 3; S → 3
(D) P → 2; Q → 1; R → 4; S → 1

44. Let four curves C_1, C_2, C_3 and C_4 in the Argand plane be defined as below
C_1 is $z^2 + \bar{z}^2 - z - \bar{z} = 2|z|^2$; C_2 is $(1+i)z + (2-i)\bar{z} + (3+i) = 0$;
C_3 is $|(1+i)z + (1-i)\bar{z} + 5| = 2|z - 5 - 2i|$; C_4 is $|(3+4i)z + (3-4i)\bar{z} + 6| = 20|z - 3 - 3i|$

LIST - I		LIST - II	
P.	Shape defined by C_1 is	1.	point circle
Q.	Shape defined by C_2 is	2.	circle
R.	Shape defined by C_3 is	3.	ellipse
S.	Shape defined by C_4 is	4.	hyperbola
		5.	parabola
		6.	pair of straight lines

The correct option is:
(A) P → 2; Q → 1; R → 5; S → 3
(B) P → 2; Q → 6; R → 4; S → 6
(C) P → 5; Q → 6; R → 5; S → 3
(D) P → 5; Q → 1; R → 4; S → 6

45. Consider four conics defined as follows
$C_1 : \frac{x^2}{10} - \frac{y^2}{4} = 1$; $C_2 : \frac{x^2}{3} + \frac{y^2}{4} = 1$; $C_3 : \frac{y^2}{2} - x^2 = 1$; $C_4 : x^2 = 16y$

LIST - I		LIST - II	
P.	One of the tangents to C_1 is	1.	y = 2x + 9
Q.	One of the tangents to C_2 is	2.	y = 2x + 7
R.	One of the normals to C_3 is	3.	y = 2x + 4
S.	One of the normals to C_4 is	4.	$y = 2x + \frac{3}{\sqrt{5}}$
		5.	y = 2x + 6
		6.	$y = 2x + \frac{6}{\sqrt{7}}$

The correct option is:
(A) P → 5; Q → 3; R → 6; S → 1
(B) P → 2; Q → 4; R → 1; S → 3
(C) P → 5; Q → 3; R → 4; S → 1
(D) P → 5; Q → 1; R → 6; S → 3

46. Match the following List-I with List-II

LIST - I		LIST - II	
P.	$\sum_{k=1}^{\infty}\frac{1}{2^k}\tan\frac{\pi}{2^{k+2}}$	1.	$\ln 2-\frac{1}{2}$
Q.	$\sum_{k=1}^{\infty}\cot^{-1}\left(k^2+\frac{3}{4}\right)$	2.	$\tan^{-1}\frac{1}{2}$
R.	$\sum_{k=1}^{\infty}\frac{1}{(2k-1)2k(2k+1)}$	3.	$\frac{\pi}{2}$
S.	$\sum_{k=1}^{\infty}\frac{1}{(4k+1)(4k-1)}$	4.	$\frac{1}{2}-\frac{\pi}{8}$
		5.	$\frac{4}{\pi}-1$
		6.	$\tan^{-1}2$

The correct option is:

(A) P → 5; Q → 6; R → 1; S → 4
(B) P → 5; Q → 3; R → 3; S → 4
(C) P → 4; Q → 6; R → 1; S → 5
(D) P → 4; Q → 2; R → 6; S → 1

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. The number of parabola(s) that can be drawn, with $x=\sqrt{3}y$ as directrix and passing through both the points $\left(4\sqrt{3}, 2\right)$ and (2, 0), is equal to

48. In a given circle if n chords are drawn, where $n \geq 2$, such that no three chords are concurrent inside the circle and m is the number of points of intersection of the chords inside the circle, then the number of line segments obtained by dividing the chords through these m points of intersection is f(n, m). Then $\left[\frac{f(8,8)}{f(6,3)}\right]$ is equal to (where [.] denotes the greatest integer function)

49. The area enclosed by the curve $(x^2 + y^2)^3 = 16x^2$ is equal to

50. If $L=\lim_{x\to 0}\frac{1}{x^5}\left(\ln\left(\frac{1+x}{1-x}\right)-2\tan x\right)$, then $\frac{1}{L}$ is equal to

51. Let C be the curve $\left|z+\frac{2}{z}\right|=2$ on the Argand plane. If a and b represent the maximum and minimum modulus of the complex numbers on C, then $a^2 + b^2$ is equal to

52. If $I_n = \int_0^{\pi/2} \cos^n x \cos nx\,dx$, then $\sqrt{\frac{I_4}{I_8}}$ is equal to

53. The minimum distance of the curve $|z - (2 + 4i)| - |z - (2 - 4i)| = 6$ from the point $2 + i$ in the Argand plane is equal to

54. Let $a_1a_2a_3a_4$ be the decimal representation of a 4-digit number. This number is said to be "non-adjacent" if for any pair (i, j), where $1 \le i, j \le 4$ and $i \ne j$, $|a_i - a_j| > 1$. Let P be the probability that a 4-digit number chosen randomly is "non-adjacent", then P is equal to _____